

2-Day Trial: Fermi Simulation Lab Partner

Candidate Brief | Two-Day Work Trial

1) Overview

Fermi's Simulation Lab contains a large catalogue of constrained, interactive science simulations. Different simulations allow learners to explore different experiments by changing conditions and observing what happens.

For this trial, build an AI Lab Partner that helps a learner use the Fermi Simulation Lab to investigate a scientific question. The product should be able to identify a useful simulation, conduct a meaningful experiment through the real lab, and discuss the observed result with the learner.

This is not a project to build another science chatbot or create new simulations. The Simulation Lab already exists. Your task is to determine how AI can make it a more effective learning experience.

2) What We Are Evaluating

- Can you understand an unfamiliar existing product and turn it into a useful AI capability?
- Can you break an open-ended problem into a workable end-to-end system?
- Can you use AI tools and web research to learn unfamiliar techniques and make them work together?
- Can you make and explain sensible technical and product decisions within two days?
- Can you distinguish what the AI knows generally from what was actually observed in the simulation?
- Can you build an evaluation system, find real failures, and improve the product using evidence?

Prior experience with this exact problem is not required. We are evaluating how effectively you learn and build with the tools available today.

3) Input and Freedom

You will receive:

- Access to the existing Fermi Simulation Lab and its simulation catalogue.
- A few example learner questions.
- A fixed Fermi-provided budget for AI API usage.

You may use third-party AI APIs, open-source libraries, AI coding tools, and web research. You may choose any models, services, programming languages, frameworks, or interface that you believe fit the problem.

You will not receive a completed agent, a prescribed integration, or instructions to use a particular framework. Deciding how your product works with the existing lab is part of the trial.

4) Minimum Product Outcome

The submission must be a runnable conversational product that genuinely uses the Fermi Simulation Lab. At minimum:

- Given a learner's scientific question, the product can choose a relevant simulation from multiple available simulations.
- It can use the selected simulation to conduct at least one real experiment and obtain an observable result.
- It can discuss that experiment and result with the learner across a follow-up interaction.
- It does not claim to have performed actions or observed results that did not occur in the Simulation Lab.

The prototype does not need to support the entire catalogue. It should demonstrate simulation selection across a representative set of at least three simulations, with one complete experiment shown end to end.

Beyond this minimum, you must decide which learner problems matter most, what the Lab Partner should do, and how the system should work.

5) Example Learner Experiences

These examples illustrate the intended product experience; they are not an implementation specification.

Explore a scientific question

Learner: *Does a heavier pendulum swing more slowly?*

The Lab Partner could find a suitable pendulum simulation, use it to investigate the question, and help the learner interpret what happened.

Diagnose an unexpected result

Learner: *I increased the battery voltage, but the current is still zero. Is the simulation broken?*

The Lab Partner should use the actual experiment state and result before deciding whether the simulation is broken or another condition explains the observation.

Challenge a conclusion

Learner: *I changed the mass and the length, and the period increased. This proves that heavier pendulums are slower.*

The Lab Partner should help the learner judge whether the experiment supports that conclusion and, if needed, conduct a better experiment.

Choose the right lab

Learner: *I want to understand why a straw looks bent in water. Can we investigate that?*

The Lab Partner could find a relevant optics simulation, conduct an experiment, and connect the observed result back to the learner's question.

6) Deliverables

- Runnable source code.
- A short README with setup and one-command run instructions.
- A short product note defining the intended learner, the problem you chose to solve, and the important decisions you made.
- A working prototype using the existing Fermi Simulation Lab.
- Saved traces connecting learner requests, simulation interactions, observed results, and product responses.
- A runnable evaluation system, including its evaluation cases and a one-command way to execute it against your product.

- A brief EVAL.md explaining the evaluation system, success criteria, results, raw examples, failure analysis, and measured improvement.
- A short demo of the working product.

7) Evaluation Requirement

You must build an evaluation system for the product. Evaluation is not an optional write-up after implementation, and EVAL.md alone is not the evaluation system.

You must:

1. Define what success means for the Lab Partner you chose to build.
2. Create a varied evaluation set of at least 10 cases spanning at least three simulations.
3. Implement a repeatable runner that executes those cases and preserves the learner input, selected simulation, simulation interaction, observed result, and product response.
4. Run your initial system and preserve the raw results.
5. Manually inspect concrete successes and failures.
6. Make one meaningful improvement based on the observed failures.
7. Re-run the same evaluation and explain what improved, regressed, or remained unresolved.

Fermi will also evaluate the submitted system on unseen learner questions and simulations. We will inspect the real Simulation Lab interactions and outputs, not only the scores reported in your README.

8) Grading Rubric (100 Points)

Category	What we are looking for	Weight
Problem Framing and Product Judgment	A valuable learner problem, sensible scope, and coherent Lab Partner experience	15
AI Engineering and System Design	Effective use of models and tools, sound technical decisions, complete end-to-end execution, and justified trade-offs	25
Simulation Use and Trustworthiness	Relevant simulation selection, meaningful real experiments, faithful use of observed state and results, and no fabricated actions	25
Evaluation System and Improvement	A runnable evaluation system, varied cases, raw interaction evidence, honest failure analysis, and measured baseline-to-improvement comparison	25
Code Quality and Runability	Reproducible setup, working commands, understandable code, and clear communication	10

A polished chatbot cannot compensate for a product that does not genuinely use the Fermi Simulation Lab. A scientifically correct explanation cannot compensate for invented simulation actions or results.

9) Non-Goals

- Building, generating, or redesigning the Fermi Simulation Lab.
- Supporting every simulation in the catalogue.
- Training a model from scratch.
- Building a complete general-purpose AI tutor.

- Production deployment, authentication, billing, or multi-user infrastructure.
- A highly polished frontend, voice interface, or avatar.

10) Timebox

Plan for **2 days**. We do not expect production-wide simulation coverage. We expect a thoughtful product decision, one complete working journey through the real lab, evidence showing where the system succeeds and fails, and one justified improvement.